# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

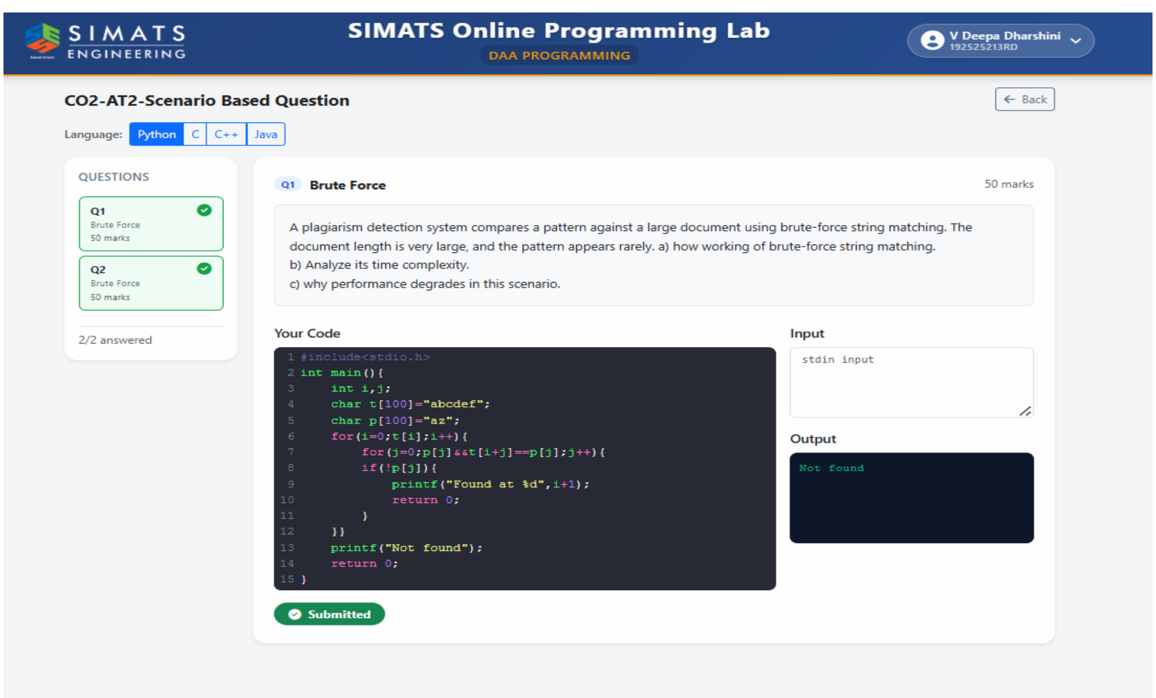
| | | | |
|---|---|---|---|
| **Course Code:** | CSA0621 | **Course Title:** | Design and Analysis of Algorithms |
| **CO Assessed:** | CO2 – Manipulation (BL1, BL2, BL3) | **Assessment:** | Assessment Tool 2 – Scenario Based Assessment |
| **Weightage:** | 20% of CIA | **Total Marks:** | 100 |
| **CO2 Statement:** | *Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)* | | |
| **Student Name:** | V Deepa Dharshini | **Reg. No.:** | 192525213 |
| **Section / Batch:** | B tech AIML(2nd year) | **Date:** | 01-08-2026 |

**Instructions to Students**

- Answer BOTH scenario-based questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

| Question Type | Focus Area | Questions | Marks |
|---|---|---|---|
| **Scenario Based Assessment** | Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems | Q1 - Q2 | 100 (2 × 50) |
| | **GRAND TOTAL:** | **100 Marks** | |

## SECTION A – SCENARIO BASED ASSESSMENT



SIMATS ENGINEERING

## SIMATS Online Programming Lab

DAA PROGRAMMING

V Deepa Dharshini
192525213RD

### CO2-AT2-Scenario Based Question

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Language: Python | C | C++ | Java

QUESTIONS

Q1
Brute Force
50 marks

Q2
Brute Force
50 marks

2/2 answered

Q2 **Brute Force** 50 marks

A student database stores records unsorted due to continuous updates. The system retrieves student details using sequential search.
a) Implement how Linear Search works here.
b) Analyze its efficiency in different cases.

Your Code

```c
#include<stdio.h>
int main(){
    int a[100],n,key,i,found=0;
    printf("Enter number of students:");
    scanf("%d",&n);
    printf("Enter students IDs:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    printf("Enter ID to search:\n");
    scanf("%d",&key);
    for(i=0;i<n;i++){
        if(a[i]==key){
            printf("Student found at position %d",i+1);
            found=1;
            break;
        }
    }
    if(!found)
        printf("Student not found");
    return 0;
}
```

Input

```
5
101 102 103 104 105
103
```

Output

```
Enter number of students:Enter
students IDs:
Enter ID to search:
Student found at position 3
```

Submitted

## Code of Conduct

*I V Deepa Dharshini (192525213) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____________

## RUBRICS FOR CALCULATION

| Criteria | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Problem Understanding & Context Analysis | 25 | Thoroughly understands the scenario with accurate interpretation of all requirements | Clearly understands the scenario with minor gaps | Basic understanding; some aspects of the scenario unclear | Poor understanding or misinterpretation of the scenario |
| Application of Concepts | 30 | Applies appropriate concepts effectively to solve complex real-world problems | Applies relevant concepts correctly with minor errors | Applies basic concepts but with limited accuracy | Fails to apply appropriate concepts |
| Analytical & Decision-Making Skills | 20 | Demonstrates strong analysis and selects optimal solutions with justification | Good analysis with reasonable solutions | Limited analysis; solutions lack justification | Poor or no analysis; incorrect decisions |
| Solution Design & Approach | 15 | Well-structured, innovative, and feasible solution approach | Logical and structured solution with minor gaps | Basic solution with limited structure or feasibility | Unclear or incorrect solution approach |
| Clarity, Justification & Presentation | 10 | Clear, well-justified explanation with strong reasoning | Clear explanation with some justification | Limited clarity and weak justification | Poorly explained with no justification |

**Marks Evaluation:**

| Question No. | Problem Understanding & Context Analysis (25%) | Application of Concepts (30%) | Analytical & Decision-Making Skills (20%) | Solution Design & Approach (15%) | Clarity, Justification & Presentation (10%) | Total Marks (Each – 50) |
|---|---|---|---|---|---|---|
| **Q1** | | | | | | |
| **Q2** | | | | | | |
| | | | | | **Overall Total (100)** | |

| Faculty In-charge | Course Coordinator | Head of Department |
|---|---|---|
| | | |
| Signature: ________________ | Signature: ________________ | Signature: ________________ |
| Date: ____________________ | Date: ____________________ | Date: ____________________ |